Ordinary Simultaneous Differential Equation

Let two ordinary simultaneous differential are:

$$f_1(D)x + g_1(D)y = R_1$$

& $f_2(D)x + g_2(D)y = R_2$

They can be solved by two methods: Elimination and Differential Method.

By Elimination Method

$$\cancel{f_1(D).f_2(D)x} + f_1(D).g_2(D)y = f_1(D)R_2$$

$$\underset{(-)}{} \cancel{f_2(D).f_1(D)x} \underset{(-)}{+} f_2(D).g_1(D)y \underset{(-)}{=} f_2(D)R_1$$

$$f_1(D).g_2(D)y - f_2(D).g_1(D)y = f_1(D)R_2 - f_2(D)R_1$$

Solve the simultaneous equations

1. $\dfrac{dx}{dt} - 7x + y = 0$

$\dfrac{dy}{dt} - 2x - 5y = 0$

Sol^n:

The given eq^{ns}. can be written as,

$$Dx - 7x + y = 0$$

$$\& \quad Dy - 2x - 5y = 0$$

$$\Rightarrow (D-7)x + y = 0$$

$$(D-5)y - 2x = 0$$

$$\Rightarrow (D-7)x + y = 0 \quad \text{——— (1)}$$

$$2x - (D-5)y = 0 \quad \text{——— (2)}$$

Multiplying eq^n. (2) by $(D-7)$ and eq^n. (1) by 2,

$$\cancel{2(D-7)x} - (D-7)(D-5)y = 0$$
$$\cancel{2(D-7)x} + 2y = 0$$
$$(-) \qquad (-)$$

$$(D-7)(D-5)y + 2y = 0$$

Date

$$\Rightarrow [(D-7)(D-5)+2]y = 0$$

The A.E. is,

~~(y-7)(y-5)~~

$$(m-7)(m-5)+2=0$$
$$\Rightarrow m^2-7m-5m+35+2=0$$
$$\Rightarrow m^2-12m+37=0$$

$$m = \frac{-(-12) \pm \sqrt{(-12)^2-4\times 1\times 37}}{2\times 1} = 6 \pm i$$

$$\therefore C.F. = y = e^{6t}(C_1 \cos t + C_2 \sin t) \quad \text{---(3)}$$

Now,

$$\frac{dy}{dt} = \frac{d}{dt}[e^{6t}(C_1 \cos t + C_2 \sin t)]$$

$$\Rightarrow \frac{dy}{dt} = e^{6t}(-C_1 \sin t + C_2 \cos t) + 6e^{6t}(C_1 \cos t + C_2 \sin t)$$

$$\Rightarrow \frac{dy}{dt} = e^{6t}[(6C_2 - C_1)\sin t + (6C_1 + C_2)\cos t]$$

Then,

$$x = \frac{1}{2}\left(\frac{dy}{dt} - 5y\right)$$

$$= \frac{1}{2}\left\{e^{6t}\left[(6C_2 - C_1)\sin t + (6C_1 + C_2)\cos t - 5(C_1 \cos t + C_2 \sin t)\right]\right\}$$

$$\therefore x = \frac{1}{2} e^{6t}\left[(C_1+C_2)\cos t + (C_2-C_1)\sin t\right] \quad \text{—(4)}$$

Hence, the equations (3) & (4) gives the solution of the given simultaneous differential equation.

2. $\dfrac{dx}{dt} + 4x + 3y = t$

$\dfrac{dy}{dt} + 2x + 5y = e^t$

Sol^n:

The given eq$^{ns.}$ can be written as,

$$Dx + 4x + 3y = t$$
$$Dy + 2x + 5y = e^t$$

$$\Rightarrow (D+4)x + 3y = t \quad \text{——(1)}$$
$$2x + (D+5)y = e^t \quad \text{——(2)}$$

Multiplying eq^n. (2) by $(D+4)$ and eq^n. (1) by 2,

$$\begin{array}{rcl} 2(D+4)x + (D+4)(D+5)y & = & e^t(D+4) \\ 2(D+4)x + 6y & = & 2t \\ (-) \qquad\quad (-) & & (-) \\ \hline (D+4)(D+5)y - 6y & = & (D+4)e^t - 2t \end{array}$$

$$\Rightarrow [D^2+9D+20-6]y = e^t + 4e^t - 2t$$

$$\Rightarrow [D^2+9D+14]y = 5e^t - 2t$$

The A.E. is,

$$m^2 + 9m + 14 = 0$$

$$\Rightarrow m^2 + 7m + 2m + 14 = 0$$

$$\Rightarrow m(m+7) + 2(m+7) = 0$$

$$\Rightarrow (m+2)(m+7) = 0$$

$$\Rightarrow m = -2, -7$$

$$C.F. = C_1 e^{-2t} + C_2 e^{-7t}$$

Now,

$$P.I. = \frac{1}{(D^2+9D+14)} \cdot (5e^t - 2t)$$

$$= 5 \cdot \frac{1}{(D^2+9D+14)} \cdot e^t - 2 \frac{1}{(D^2+9D+14)} \cdot t$$

$$= 5e^t \frac{1}{(1^2+9\times1+14)} - \frac{2}{14} \cdot \frac{1}{\left(1 + \frac{D^2+9D}{14}\right)} \cdot t$$

$$= \frac{5e^t}{24} - \frac{1}{7}\left[1 + \frac{D^2+9D}{14}\right]^{-1} \cdot t$$

$$= \frac{5e^t}{24} - \frac{1}{7}\left\{1 - \frac{D^2+9D}{14} + \dots\right\} \cdot t$$

Removing higher order of D,

$$P.I. = \frac{5e^t}{24} - \frac{1}{7}\left[t - \frac{D^2(t) + 9D(t)}{14}\right]$$

$$= \frac{5e^t}{24} - \frac{1}{7}\left(t - \frac{9}{14}\right)$$

$$= \frac{5e^t}{24} - \frac{1}{7}t + \frac{9}{98}$$

$\therefore y = C.F. + P.I.$

$$\Rightarrow y = C_1 e^{-2t} + C_2 e^{-7t} + \frac{5e^t}{24} - \frac{1}{7}.t + \frac{9}{98} \quad \text{—(3)}$$

Now,

$\cancel{\frac{dy}{dt} + 2x + 5y = e^t}$

$\cancel{\Rightarrow x =}$

$$\frac{dy}{dt} = -2C_1e^{-2t} - 7C_2e^{-7t} + \frac{5}{24}.e^t - \frac{1}{7}$$

Also,

$$x = -\frac{1}{2}\left(\frac{dy}{dt} + 5y\right) + e^t$$

$$= -\frac{1}{2}\left[-2C_1e^{-2t} - 7C_2e^{-7t} + \frac{5}{24}e^t - \frac{1}{7} + 5C_1e^{-2t} + 5C_2e^{-7t} + \frac{25}{24}e^t - \frac{5}{7}t + \frac{45}{98}\right] + e^t$$

Date

3. $\dfrac{dx}{dt} = 3x + 2y$

$\dfrac{dy}{dt} = 5x + 3y$

Soln:

The given equations can be written as,

$$Dx - 3x - 2y = 0$$
$$Dy - 5x - 3y = 0$$

$$\Rightarrow (D-3)x - 2y = 0$$
$$-5x + (D-3)y = 0$$

$$\Rightarrow (D-3)x - 2y = 0 \quad \text{——} \quad (1)$$
$$5x - (D-3)y = 0 \quad \text{——} \quad (2)$$

Multiplying eqn. (2) by $(D-3)$ & eqn. (1) by 5,

$$\cancel{5(D-3)x} - (D-3)(D-3)y = 0$$
$$\underset{(-)}{\cancel{5(D-3)x}} \; \underset{(+)}{-} \; 10y = 0$$

$$-(D-3)^2 y + 10y = 0$$

$$\Rightarrow [(D-3)^2 - 10]y = 0$$

The A.E. is,

$$(m-3)^2 - 10 = 0$$

$$\Rightarrow m^2 - 6m + 9 - 10 = 0$$

$$\Rightarrow m^2 - 6m - 1 = 0$$

$$m = \frac{-(-6) \pm \sqrt{(-6)^2 - 4 \times 1 \times (-1)}}{2 \times 1}$$

$$= \frac{6 \pm \sqrt{40}}{2}$$

$$= 3 \pm \frac{2\sqrt{10}}{2}$$

$$\Rightarrow m = 3 \pm \sqrt{10}$$

$\therefore$ C.F $y = e^{3t}(C_1 \cosh t\sqrt{10} + C_2 \sinh t\sqrt{10})$ —(3)

Now,

$$\frac{dy}{dt} = e^{3t}(C_1\sqrt{10} \sinh t\sqrt{10} + C_2\sqrt{10} \cosh t\sqrt{10}) + 3e^{3t}(C_1 \cosh t\sqrt{10} + C_2 \sinh t\sqrt{10})$$

$$= e^{3t}\left\{(3C_1 + \sqrt{10}C_2) \cosh t\sqrt{10} + (3C_2 + \sqrt{10}C_1) \sinh t\sqrt{10}\right\}$$

Then,

$$x = \frac{1}{5}\left\{\frac{dy}{dt} - 3y\right\}$$

$$= \frac{1}{5}\left\{e^{3t}\left\{(3C_1 + \sqrt{10}C_2)\cosh t\sqrt{10} + (3C_2 + \sqrt{10}C_1)\sinh t\sqrt{10} - 3C_1 \cosh t\sqrt{10} - 3C_2 \sinh t\sqrt{10}\right\}\right\}$$

Page

Date

$$\therefore x = \frac{1}{5} e^{3t} \{\sqrt{10}\, C_2 \cosh t\sqrt{10} + \sqrt{10}\, C_1 \sinh t\sqrt{10}\}$$

$$= \frac{\sqrt{10}}{5} \cdot e^{3t} \{C_2 \cosh t\sqrt{10} + C_1 \sinh t\sqrt{10}\} \quad \text{---} (4$$

Hence, eqⁿ. (3) & eqⁿ. (4) gives the solution.

4. $\frac{dy}{dx} + y = z + e^x$

$\frac{dz}{dx} + z = y + e^x$

Solⁿ:

The given equations can be written as,

$$Dy + y - z = e^x$$
$$Dz + z - y = e^x$$

$$\Rightarrow (D+1)y - z = e^x \quad \text{---} (1)$$
$$y - (D+1)z = e^x \quad \text{---} (2)$$

Multiplying eqⁿ. (2) by $(D+1)$ and eqⁿ. (1) by 1,

$$\cancel{(D+1)y} - (D+1)^2 z = (D+1)e^x$$
$$\cancel{(D+1)y} - z = e^x$$
$$(-) \quad (+) \quad (-)$$

$$-(D+1)^2 z + z = (D+1)e^x - e^x$$
$$\Rightarrow (-D^2 - 2D - 1 + 1)z = e^x$$

$$\Rightarrow (-D^2-2D)z = e^x$$

$$\cancel{\Rightarrow (D^2+2D}$$

$$\Rightarrow -D(D+2)z = e^x$$

$$\Rightarrow D(D+2)z = -e^x$$

The A.E. is,

$$m(m+2) = 0$$

$$\Rightarrow m = 0, -2$$

$$\therefore C.F. = C_1 + C_2 e^{-2x}$$

$$P.I. = \frac{1}{(D^2+2D)} \cdot (-e^x)$$

$$= -e^x \cdot \frac{1}{1^2+2\times 1}$$

$$= \frac{-e^x}{3}$$

$$\therefore z = C.F. + P.I. = C_1 + C_2 e^{-2x} - \frac{e^x}{3} \quad \text{---(3)}$$

$$\frac{dz}{dx} = -2C_2 e^{-2x} - \frac{e^x}{3}$$

$$y = \frac{dz}{dx} + z - e^x = -2C_2 e^{-2x} - \frac{e^x}{3} + C_1 + C_2 e^{-2x}$$

$$\frac{-e^x}{3} - e^x$$

Date

$$\therefore y = C_1 - C_2e^{-2x} - \frac{5e^x}{3} \quad \text{—— (4)}$$

Hence, eq^{ns} (3) & (4) gives the solution of the given simultaneous differential equation.